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Lung Nodule Prognostics

– DS - Intelligent Modeling report –

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2025-2026

Abstract

Lung cancer screening with low-dose computed tomography (CT) detects large numbers of *indeterminate* pulmonary nodules whose clinical fate is hard to predict: some stay harmless for years, while others progress into aggressive cancer. Today this risk stratification rests on the subjective visual judgement of radiologists, which varies between readers and can lead either to overtreatment of benign nodules or to dangerous delays for malignant ones. This report asks whether an automated, *intelligent* model can support that decision by learning several biologically related properties of a nodule at once. We propose *MultiTaskCrossTalkNet*, a multi-task convolutional neural network that jointly predicts a nodule’s radiological subtype (ground-glass, part-solid, solid) and its malignancy risk (benign vs. malignant). Instead of rigidly sharing every layer, the network keeps one convolutional branch per task and links them through learnable *cross-talk* modules—a cross-stitch unit and a co-attention unit—so each task can decide how much to borrow from the other. We benchmark this design against single-task baselines that share no information.

All experiments use 508 nodule crops from the public LIDC-IDRI dataset, converted to Hounsfield units, cropped around the annotated region of interest and resized to 128×128 pixels. A third clinical task—two-year progression—was part of the original design but had to be dropped, because the progression labels in the clinical records could not be reliably linked to the imaging annotations. Contrary to our hypothesis, cross-talk did *not* consistently outperform single-task learning under the conditions tested. The subtype task collapsed to the majority “solid” class because of extreme class imbalance (97 of 101 test samples were solid), and malignancy prediction gained nothing reliable from information sharing. We attribute this to data quality and class distribution rather than to the architecture itself, and we identify class-balanced sampling, volumetric (3D) inputs, and a corrected progression linkage as the conditions under which the multi-task hypothesis is most likely to pay off. We report this honest negative result as a useful signal for others applying multi-task learning to small, imbalanced medical-imaging datasets.

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Chapter 1

Introduction

1.1 What? Why? How?

Lung cancer screening saves lives, but it frequently reveals “indeterminate” lung nodules that could be harmless and dormant, or aggressive and deadly. The scientific problem we address is the accurate, automated risk stratification of these uncertain nodules from low-dose CT scans.

Currently, deciding what to do with an indeterminate nodule relies heavily on a radiologist’s subjective visual assessment of its size, shape, and physical texture. Because this process is vulnerable to human error and inter-reader variability, it often leads to two devastating extremes: overtreatment, where healthy patients are subjected to invasive, expensive, and stressful biopsies, or undertreatment, where malignant tumors are missed until it is too late. The underlying visual data is simply too complex and high-dimensional for the human eye to process with perfect consistency.

To address this, we developed an intelligent algorithm that mimics the interconnected nature of clinical biology. Traditional machine learning models often treat a nodule’s physical traits (like its texture) and its ultimate cancer risk as completely separate problems. Our approach bridges this gap. We propose *MultiTaskCrossTalkNet*, a multi-task deep learning architecture that evaluates these factors simultaneously. By utilizing “cross-talk” mechanisms, the model dynamically shares information between its internal pathways, learning that a nodule’s physical composition, malignancy risk and its rate of progression are biologically linked.

1.2 Paper structure and original contribution(s)

The main contribution of this report is to present an intelligent algorithm for solving the problem of simultaneous lung nodule texture classification, malignancy risk prediction and progression classification

using a multi-task deep learning approach with cross-talk mechanisms.

The present work is structured in five chapters as follows.

The first chapter introduces the research topic, explains its importance, and presents the objectives and contributions of the thesis. It also provides an overview of the structure of the paper.

The second chapter presents the scientific problem addressed in this thesis, explicitly defining the formal inputs, outputs, and the necessity for an intelligent multi-task approach.

The third chapter presents the related work and describes existing approaches.

The fourth chapter presents the data that was used, details the proposed methodology, outlines the experiments and results, and concludes with a discussion of the model's strengths and limitations.

Finally, the last chapter summarizes the conclusions of the thesis and outlines possible future improvements and research directions.

Chapter 2

Scientific Problem

2.1 Problem description

Low-dose computed tomography (CT) screening has drastically improved the early detection of lung cancer, but at the same time it introduced a clinical dilemma: the management of indeterminate lung nodules. When a nodule is detected, its clinical trajectory is highly uncertain. It may remain harmless and slow-growing for a patient's entire life, but it could also progress into aggressive metastatic lung cancer at a fast pace.

Currently, radiologists and oncologists rely on a subjective synthesis of heterogeneous attributes, such as the nodule's radiological subtype (solid, part-solid, ground-glass), volumetric growth patterns over time, and subtle morphological textures to stratify risk. Because this manual assessment is highly variable among practitioners, it frequently leads to two adverse outcomes: overtreatment, where patients with benign nodules are subject to invasive, risky, and expensive biopsies or undertreatment, where aggressive tumors are not removed in time.

2.2 Why an Intelligent Algorithm is Mandatory

This problem requires an intelligent algorithm because the underlying data exceeds human cognitive and visual processing capacities. The human visual system cannot reliably extract tiny, complex patterns from 3D medical images. Furthermore, rule based clinical guidelines or standard linear statistical models fail because lung cancer progression depends on many complex factors.

Deep learning algorithms are uniquely capable of automatically extracting latent hierarchical representations from raw pixel and voxel data. More importantly, a multi-task algorithm is hypothesized to better capture the biological reality that a nodule's morphology, growth rate, and ultimate malignancy

are deeply interdependent phenomena, rather than isolated variables.

2.3 Solution Methods

To effectively predict nodule malignancy (T1), progression (T2), and subtype (T3) from medical images, we evaluate two different frameworks: an approach that treats each task separately, and a joint-learning method designed to capture their biological overlap. Note that while the full three-task formulation defines the intended scope of this work, T2 (progression) was excluded from the final implementation due to a data linkage constraint described in detail in Section 4.3: the clinical file containing progression labels does not share a common identifier with the DICOM XML annotation files, making it impossible to unambiguously match progression labels to individual CT crops.

2.3.1 Traditional Baseline: Single-Task Learning (STL)

The traditional baseline approach utilizes Single-Task Learning, where separate networks are trained independently for malignancy, progression, and subtype. The primary advantage of these models is that they are straightforward to implement and optimize. Hyperparameter tuning is relatively simple, and the loss landscape for a single objective is generally well behaved. However, a major disadvantage is that STL completely ignores the biological correlations between the tasks. It forces the models to learn redundant feature extractors from scratch for each task. This is highly inefficient and increases the risk of overfitting, which is a critical flaw when working with limited medical datasets like LIDC-IDRI or LNDb.

2.3.2 Proposed Solution: Multi-Task Learning (MTL) with Cross-Talk

The proposed solution employs Multi-Task Learning equipped with cross-talk. By jointly predicting T1, T2, and T3, MTL acts as a strong inductive bias and a natural regularizer. Shared layers allow the model to leverage larger, combined pools of information, making the learned representations far more robust. The advantage of cross-talk mechanisms, such as cross-stitch units or co-attention, is that they elevate this by dynamically controlling how information flows between task specific branches. This allows the model to autonomously determine, for example, how much a specific visual texture should influence the overall malignancy prediction. On the other hand, a significant disadvantage is that MTL architectures are difficult to tune. They are highly susceptible to negative transfer, where optimizing one task degrades the performance of another, or task domination, where the loss of a complex task overwhelms a simpler one.

2.4 Formal Problem Definition

The objective is to formulate a multi-task learning mapping function capable of joint inference over nodule-level CT imaging data.

2.4.1 Inputs

Let the input space be defined as $X \subset \mathbb{R}^{1 \times 128 \times 128}$, where each element $x \in X$ is a single-channel, two-dimensional CT crop of size 128×128 pixels centered on a radiologist-annotated nodule region of interest. Raw pixel values are converted to Hounsfield Units (HU) using the DICOM `RescaleSlope` and `RescaleIntercept` attributes, clipped to the range $[-1000, 400]$ HU, and normalized to $[0, 1]$.

2.4.2 Outputs

The MTL model learns a parameterized function $f_\theta : X \rightarrow (\hat{Y}_1, \hat{Y}_2, \hat{Y}_3)$ to predict three distinct target spaces:

- **T1 — Nodule malignancy risk:** $\hat{Y}_1 \in \{0, 1\}$, where 0 is Benign and 1 is Malignant.
- **T2 — Two-year progression probability:** $\hat{Y}_2 \in \{0, 1\}$, where 0 is Stable and 1 is Progressive.
- **T3 — Radiological subtype classification:** $\hat{Y}_3 \in \{0, 1, 2\}$, representing a multiclass prediction (0 = solid, 1 = part-solid, 2 = ground-glass).

2.4.3 Objective

The network optimizes task-specific parameters $\theta_1, \theta_2, \theta_3$, potentially alongside a foundational shared backbone θ_{shared} , and cross-talk routing parameters α (cross-stitch matrices / co-attention weights). These parameters are jointly trained to minimize a globally weighted loss function:

$$\mathcal{L}_{\text{total}} = \lambda_1 \mathcal{L}_{\text{BCE}}(\hat{Y}_1, Y_1) + \lambda_2 \mathcal{L}_{\text{BCE}}(\hat{Y}_2, Y_2) + \lambda_3 \mathcal{L}_{\text{CE}}(\hat{Y}_3, Y_3)$$

where $\mathcal{L}_{\text{BCE}}$ is Binary Cross-Entropy, $\mathcal{L}_{\text{CE}}$ is Categorical Cross-Entropy, and λ_i are dynamic weighting hyperparameters to balance task gradients.

2.5 Motivation

This problem directly impacts patient survival. An accurate, generalized model can fundamentally alter the clinical workflow for pulmonary oncology. By accurately identifying high risk nodules early, life saving treatments can start sooner. On the flip side, by confidently identifying indolent nodules,

the system reduces the immense psychological and financial cost of unnecessary biopsies and repeated radiation exposure.

From an AI perspective, translating deeply interconnected biological traits into a computational architecture is a state-of-the-art challenge. Designing cross-talk mechanisms that avoid negative transfer in noisy, high-dimensional medical data pushes the boundaries of multi-task network design.

Chapter 3

Related work

Recent advances in deep learning have significantly improved automated analysis of medical imaging data, particularly in lung cancer screening using computed tomography (CT). Lung nodules detected during low-dose CT screening present a major diagnostic challenge, as their clinical significance varies widely: some nodules remain stable for years, while others rapidly progress to malignancy. Accurate risk stratification therefore requires the integration of multiple heterogeneous sources of information, including imaging characteristics, volumetric growth patterns, and patient clinical data.

Multi-task learning (MTL) has been applied to improve lung nodule classification in CT images. The authors of [9] propose a multi-task CNN (MT-CNN) that distinguishes malignant from benign nodules using two tasks: a main classification task and an auxiliary image reconstruction task. The model extracts nine 2D views from each 3D nodule and processes each view with a shared network followed by two branches: one for classification and one for reconstruction. The final prediction is obtained by a weighted fusion of the nine view-specific models, allowing the network to capture 3D spatial information while maintaining lower computational complexity.

The method is evaluated on two publicly available datasets, LIDC-IDRI and LUNA-16. LIDC-IDRI contains 1,018 annotated CT scans with radiologist-provided malignancy scores, while LUNA-16 is a subset of LIDC-IDRI designed for nodule detection and classification tasks.

The proposed MT-CNN achieves strong performance compared to state-of-the-art approaches, reaching an AUC of 97.3% on LUNA-16 and 95.59% on LIDC-IDRI, along with a low false positive rate of 3.2%. These results demonstrate that incorporating an auxiliary reconstruction task helps preserve fine-grained image features and improves classification accuracy.

Our work differs in that we address a prognostic setting with multiple clinically meaningful tasks: malignancy risk prediction, two-year progression probability, and radiological subtype classification. Instead of using reconstruction as an auxiliary task, our model jointly learns these related clinical

endpoints. Our approach is more informative for clinical decision support because it predicts several complementary outcomes simultaneously and analyzes their interactions, providing richer insights than a single malignancy classification task.

MTL has also been explored to improve both prediction accuracy and interpretability in lung nodule analysis. Wang et al.[8] propose IMAL-Net, an interpretable multi-task attention network for screening invasive lung adenocarcinoma from CT images. Their method jointly performs two tasks: lung nodule segmentation and invasiveness classification, using two sub-ResNet networks connected through a prior-attention mechanism. The segmentation branch provides spatial priors that guide the classification branch, while radiomic features extracted from segmentation masks are fused with deep semantic features to improve both performance and interpretability.

The model is trained and evaluated on a large, non-public clinical dataset consisting of 1,626 patients collected from two hospitals, with biopsy-confirmed labels, and an additional external validation set of 200 nodules. This makes the dataset clinically reliable but not publicly available.

IMAL-Net achieves strong performance, reaching an AUC of 93.8% and a recall of 93.8% in identifying invasive lung adenocarcinoma. The results show that combining segmentation, attention mechanisms, and radiomic-semantic feature fusion improves both classification accuracy and model interpretability.

In our work, we focus on lung nodule prognostics with three prediction tasks: malignancy risk, two-year progression probability, and radiological subtype classification. Instead of combining segmentation and classification only, our model learns multiple clinically meaningful endpoints simultaneously and explicitly models interactions between them. Our approach provides richer clinical information because it predicts several complementary outcomes within a single framework. By jointly modeling malignancy, progression risk, and radiological subtype, the model can capture relationships between these factors and support more comprehensive clinical decision making.

Multi-task learning has also been used to jointly exploit the relationships between different lung nodule analysis tasks. Wang et al.[7] propose DeepLN, a multi-task deep learning framework that simultaneously predicts imaging characteristics, malignancy, and pathological subtypes from CT scans. The model is based on a 3D-ResNet backbone and incorporates auxiliary tasks, such as nodule density and morphological feature classification, to guide the main tasks and improve feature representation.

The model is trained on a large retrospective clinical dataset of 8,950 pulmonary nodules from 5,823 patients, with pathology-confirmed labels serving as the gold standard. This dataset is not publicly available but provides strong clinical reliability.

DeepLN demonstrates strong performance across multiple tasks. For malignancy prediction, it achieves an AUC of 0.8503 on the test set, while for pathological subtype classification, it reaches AUCs of 0.8675 for adenocarcinoma and 0.8792 for squamous cell carcinoma. Additionally, the inclusion of auxiliary imaging features improves performance compared to single-task models, highlighting the benefit of multi-task learning.

In contrast, our work focuses on lung nodule prognostics rather than combining imaging feature prediction with classification tasks. Our model jointly predicts malignancy risk, two-year progression probability, and radiological subtype, which represent clinically meaningful endpoints rather than intermediate imaging features. By modeling these related outcomes together, our approach provides more comprehensive clinical decision support and captures relationships between biological and radiological factors.

Another multi-task approach is proposed by Ni et al.[4], who introduce a two-stage multitask U-Net framework for pulmonary nodule segmentation and malignancy risk prediction. Their method follows a coarse-to-fine strategy, where a 3D multiscale U-Net (MU-Net) first localizes the nodule region, followed by a 2.5D multiscale separable U-Net (MSU-Net) that refines segmentation and performs multi-task classification. The model jointly predicts malignancy along with radiological characteristics such as margin, texture, and calcification by sharing features between segmentation and classification branches.

The approach is evaluated on the public LIDC-IDRI dataset, using 1,561 annotated nodules with radiologist-provided labels. Experimental results show strong performance, achieving a Dice similarity coefficient of 83.04% for segmentation and an AUC of 84.3% for malignancy prediction. The results demonstrate that integrating segmentation and classification in a multitask framework improves both feature representation and predictive accuracy.

While this approach improves diagnostic performance by leveraging segmentation cues and radiological feature prediction, it focuses primarily on detection and characterization tasks. In contrast, our work addresses a broader prognostic setting by jointly modeling malignancy risk, progression probability, and radiological subtype. This formulation enables the model to capture dependencies between clinically meaningful outcomes and provide richer insights into lung nodule behavior.

Radiomics-based approaches have also been proposed to predict the growth behavior of pulmonary nodules. Sun et al.[5] develop a CT-based radiomics model to predict the growth or long-term stability of pulmonary ground-glass nodules (GGNs). Their method involves manual segmentation of nodules followed by extraction of high-dimensional radiomic features (1218 features), including intensity, shape,

and texture descriptors. Feature selection is performed using mRMR and LASSO, and a radiomics signature is constructed via logistic regression.

The model is trained and validated on a dataset of 253 patients with 1115 CT images, split into training (70%) and validation (30%) sets. A combined nomogram integrating radiomics features with clinical factors (age, size, and location) achieves the best predictive performance, with AUC values of 0.843 in the training set and 0.824 in the validation set. The radiomics model significantly outperforms traditional clinical features, demonstrating the advantage of capturing tumor heterogeneity through quantitative imaging descriptors.

However, this approach focuses on a single prediction task, namely distinguishing between growing and stable nodules. In contrast, our work adopts a multi-task learning framework that simultaneously predicts malignancy risk, progression probability, and radiological subtype. By jointly modeling multiple clinically relevant outcomes, our approach captures interdependencies between different aspects of lung nodule behavior and provides more comprehensive prognostic information for clinical decision making.

Another related approach focuses on the automatic categorization of pulmonary nodules based on their radiological characteristics. Tu et al.[6] propose a convolutional neural network (CNN)-based computer-aided diagnosis (CADx) system for the automatic categorization of solid, part-solid, and non-solid pulmonary nodules in CT images. Their method operates directly on two-dimensional regions of interest (ROIs) without requiring explicit nodule segmentation, allowing the model to learn hierarchical features directly from image context and avoid errors associated with segmentation.

The framework supports both classification (three nodule types) and regression (radiological scoring from 1 to 5), and is trained on the LIDC dataset containing annotations from multiple radiologists. Experimental results show that the CNN significantly outperforms traditional histogram-based methods and achieves substantial agreement with expert radiologists, with Cohen's kappa values around 0.74.

While this work focuses on improving automated categorization and scoring of nodule types, it primarily addresses a single aspect of nodule characterization. In contrast, our approach extends this idea by incorporating radiological subtype classification as one of several tasks within a multi-task learning framework. By jointly predicting radiological subtype together with malignancy risk and progression probability, our model leverages shared representations across tasks and provides a more comprehensive characterization of lung nodules for clinical decision support.

A different multi-task strategy for lung nodule analysis is proposed in MTMR-Net [3]. This model

jointly performs benign–malignant classification and regression of multiple radiological semantic attributes, including spiculation, lobulation, sphericity, calcification, and others. The architecture is based on a shared ResNet backbone with two task-specific branches: a classification module and a regression module, where attribute features are explicitly fed into the classification branch to model a cause-and-effect relationship between semantic characteristics and malignancy. In addition, the model incorporates a Siamese network trained with a margin ranking loss to improve discrimination between ambiguous nodules by enforcing correct ordering of malignancy scores.

The method is evaluated on the publicly available LIDC-IDRI dataset, which contains 1,018 CT scans and 1,422 annotated nodules with malignancy scores and radiological attributes provided by multiple radiologists. The experiments are conducted using five-fold cross-validation to ensure robustness.

Experimental results show that MTMR-Net achieves strong performance, reaching an accuracy of 93.5%, sensitivity of 93.0%, and AUC of 0.979 for malignancy classification, outperforming several state-of-the-art methods. Additionally, the model achieves lower absolute error in attribute score regression compared to traditional methods such as LASSO and elastic net, demonstrating the effectiveness of jointly modeling classification and semantic attributes.

In contrast, our work focuses on lung nodule prognostics with multiple clinical endpoints rather than malignancy classification alone. Besides predicting malignancy risk, our model also estimates two-year progression probability and radiological subtype, capturing complementary aspects of nodule behavior over time. This approach provides richer clinical insight by modeling interactions between several prognostic outcomes and better reflects the biological complexity of lung nodules for clinical decision making.

Recent work has also explored the integration of domain-specific semantic information into deep learning models for lung nodule analysis. Dong et al. [2] propose a 3D convolutional neural network (SCCNN) that incorporates radiological semantic characteristics as auxiliary tasks within a multi-task learning framework. The model extracts multi-view 3D patches of lung nodules and processes them using a shared CNN backbone with task-specific branches for malignancy classification and semantic feature prediction. In addition, a Convolutional Block Attention Module (CBAM) is introduced to enhance feature fusion by focusing on the most informative spatial and channel-wise features.

The model is trained and evaluated on the publicly available LIDC-IDRI dataset, which contains over 1,000 CT scans with annotations from multiple radiologists, including semantic characteristics such as lobulation, spiculation, subtlety, and malignancy scores. After preprocessing and data cleaning,

1,779 nodules are used for training and evaluation.

Experimental results show that the proposed SCCNN significantly outperforms a baseline 3D CNN. The model achieves an AUC of up to 0.973 and an accuracy of 95.45% when combining semantic characteristics and attention mechanisms, compared to 89.77% accuracy for the baseline model. These results demonstrate that incorporating semantic features improves both classification performance and interpretability.

In contrast, our work focuses on lung nodule prognostics with multiple clinical endpoints. Instead of predicting malignancy alone, our model jointly estimates malignancy risk, two-year progression probability, and radiological subtype. This allows us to capture relationships between different aspects of nodule behavior and provide more comprehensive prognostic information for clinical decision making.

In addition to task-specific approaches, several studies have reviewed the broader landscape of multi-task learning in deep neural networks. Crawshaw[1] provides a comprehensive survey of multi-task learning methods and their applications across different domains. The survey discusses the main categories of MTL architectures, including hard parameter sharing, soft parameter sharing, cross-stitch networks, and attention-based task interaction mechanisms. It also highlights the benefits of MTL in improving generalization, reducing overfitting, and enabling knowledge transfer between related tasks. The survey emphasizes that MTL is particularly effective when tasks share underlying representations, which has motivated its increasing adoption in complex prediction problems such as medical image analysis.

Chapter 4

Investigated approach

4.1 Data

The primary dataset utilized in this study is the LIDC-IDRI (Lung Image Database Consortium and Image Database Resource Initiative) dataset. As a premier public resource for lung nodule analysis, it provides a comprehensive collection of thoracic CT scans, unblinded radiologist annotations, and clinical diagnostic records. Because nodules are frequently reviewed by multiple radiologists, the dataset captures inherent clinical variability and uncertainty, making it highly representative of real-world diagnostic scenarios. For this analysis, the data is divided into three interconnected components: the raw imaging data, the clinical diagnosis records and the nodule annotations.

4.1.1 Radiological Imaging Data

The imaging data consists of 3D CT scans stored as raw DICOM files. The pipeline recursively extracts files with the “CT” modality and reconstructs the full axial chest volume by sorting the 2D slices along the Z-axis (utilizing spatial attributes such as ImagePositionPatient). Each 2D slice has a standard image size of 512x512 pixels. To ensure accurate radiodensity representations, the raw pixel arrays are converted into standardized Hounsfield Units (HU) using the RescaleSlope and RescaleIntercept metadata.

4.1.2 Radiologist Nodule Annotations

Accompanying the DICOM files are XML annotation files documenting findings from radiologist reading sessions. These annotations describe the location, size, shape, internal structure, margin sharpness, and visual appearance of the identified lesions. A critical characteristic extracted for this study is the nodule texture, which indicates how dense or solid a lesion appears on a CT scan.

The original texture scores in the dataset range from 1 to 5, each corresponding to the following labels, in order: non-solid, mostly non-solid, part-solid, mostly solid, solid

To reduce noise and facilitate comparative analysis, these scores were mapped into three distinct radiological categories:

Ground-Glass (Scores 1 & 2): Lighter, less dense regions.

Part-Solid (Score 3): Lesions containing both dense and non-dense regions.

Solid (Scores 4 & 5): Denser, brighter regions.

4.1.3 Clinical Diagnosis Data

A supplementary clinical dataset provides patient and nodule-level diagnostic context across 157 patient records. This component contains variables detailing clinical outcomes, including the overall diagnosis result, tumor location, and the specific method used to confirm the diagnosis (e.g., biopsy, surgical resection, radiological follow-up, or progression). For metastatic cases, the primary tumor site is recorded across 112 unique classes (e.g., non-small cell lung cancer, melanoma, colon cancer).

4.1.4 Data Processing and Feature Engineering

To structure the data for modeling, the annotation data was aggregated into two distinct relational tables.

One is a detailed table with one row for each texture annotation, containing the following information: Patient ID, Nodule ID, Reading session index, Texture score and Texture category.

The other table is a patient-level summary table that combines all annotations for a patient and stores the following data: Total number of texture annotations, Number of ground-glass nodules, Number of part-solid nodules, Number of solid nodules, Dominant texture category.

The dominant texture category is the one that appears most often for that patient. For example, if most nodules for a patient are solid, then the patient is labeled as having a dominant solid texture pattern.

4.1.5 Machine Learning Target Variables

Predicting exact medical diagnoses across highly granular classes is challenging, particularly given the sample size constraints of the clinical dataset. To create robust predictive models, the clinical labels were simplified into three distinct binary and multiclass targets, focusing on the most critical medical distinctions:

Malignancy prediction: 0 for benign (derived from “benign/non-malignant” labels) and 1 for malignant (derived from “malignant primary lung cancer” and “malignant metastatic lesion” labels);

Progression prediction: 0 for stable (nodule remained stable for at least two years) and 1 for progressive (documented progression);

Radiological subtype: as previously mentioned: 0 for solid, 1 for part-solid and 2 for ground-glass

This makes the classification problem easier and helps the model focus on the most important medical distinction: whether the nodule is dangerous or not.

4.1.6 Exploratory Data Analysis

After extracting and inspecting the data, several properties relevant to modeling were identified.

Regarding texture distribution, solid nodules are by far the most frequent category, followed by ground-glass and part-solid. This class imbalance is consistent with clinical literature and must be accounted for during training. Furthermore, most CT series contain only one or two annotated nodules. The distribution of nodule counts per scan is heavily right-skewed, meaning a small number of scans account for a disproportionately large share of the total annotations.

Because each patient may have been reviewed by up to four radiologists in separate reading sessions, the same nodule can receive different texture scores from different readers. This inter-reader variability was partially mitigated at the patient level by applying a majority-vote approach to establish a single “dominant texture category” across all annotations.

In the diagnosis dataset, the nodule-level columns for higher-indexed nodules (e.g., nodules 4 and 5) contain a massive fraction of missing values, reflecting the fact that most patients have fewer than four annotated nodules. Additionally, the `primary_tumor_site` column was excluded as a feature; it applies specifically to metastatic cases and suffers from extreme high cardinality (containing over 110 unique free-text classes for just 157 patients), rendering it too fragmented for robust model training.

Finally, both derived classification targets exhibit class imbalance. The malignancy target is moderately skewed toward the benign class, while the progression target is severely unbalanced, with stable cases substantially outnumbering progressive ones.

Overall, the dataset is valuable because it combines CT images, radiologist annotations, and diagnosis information. This supports both exploratory analysis and predictive modeling. Furthermore, the presence of missing values, annotation variability, and multiple expert opinions makes the dataset more representative of real clinical scenarios.

4.2 Proposed Architecture

To address the problem of simultaneous texture, malignancy, and progression classification, we propose a multi-task learning architecture termed *MultiTaskCrossTalkNet*. The primary objective of this architecture is to evaluate the efficacy of simultaneously predicting a nodule’s physical texture, its immediate malignancy risk, and its long-term clinical progression. Specifically, the network relies on the hypothesis that soft parameter sharing (cross-talk) between these three highly correlated clinical tasks improves overall diagnostic performance compared to isolating them. Because physical composition, cancer risk, and growth rate are biologically linked, allowing the model to share learned feature representations should theoretically yield a more robust predictive model.

Instead of hard parameter sharing (which forces all tasks to use the exact same early-layer weights), this proposed network employs three parallel Convolutional Neural Network (CNN) branches: one dedicated to texture, one to malignancy, and one to progression. After each convolutional block, a learnable *CrossStitchUnit* linearly combines the activation maps across the parallel branches. This allows the network to dynamically learn how much information to share between the tasks at varying levels of abstraction.

The network is designed to output three predictions from its respective classification heads:

1. A 3-class target for the dominant texture category (ground-glass, part-solid, solid), represented via one-hot encoding.
2. A binary target representing the nodule’s immediate risk level (0 for benign, 1 for malignant).
3. A binary target representing the nodule’s longitudinal progression (0 for stable, 1 for progressive).

The three-task formulation above describes the full, intended design of *MultiTaskCrossTalkNet*, with one branch per task and a learnable cross-stitch matrix whose size matches the number of active tasks. As explained in the Scope and Limitations discussion of Section 4.3, the progression task (T2) could not be reliably linked to the imaging data and therefore had to be excluded from every experiment. All results reported in this chapter are consequently for the resulting two-task instantiation: a texture branch and a malignancy branch, coupled by a 2×2 cross-stitch matrix after each convolutional block.

4.3 Experiments and Results

4.3.1 Single-Task Learning (STL)

The STL baseline consists of two independently trained CNN models: one for texture classification and one for malignancy prediction, that share no parameters and receive no signal from one another.

Each model is architecturally identical to a single branch of the full *MultiTaskCrossTalkNet*, with the cross-stitch units removed and a single task-specific classification head attached directly to the final convolutional block. This design ensures that any performance difference between STL and MTL is attributable solely to the cross-talk mechanism, and not to differences in feature extraction capacity.

The same preprocessing pipeline, train/validation/test split, and AdamW optimiser were applied as in the MTL experiments, so results are directly comparable. Each model was trained to minimise its respective loss function independently: categorical cross-entropy for the texture model and binary cross-entropy for the malignancy model. The best-performing checkpoint was selected based on lowest validation loss and subsequently evaluated on the held-out test set.

4.3.1.1 Preliminary Experiment (15 samples)

An initial feasibility run was conducted on a severely restricted subset of 15 annotated samples to validate the training pipeline end-to-end before scaling up.

The malignancy STL model achieved a test accuracy of 66.67%, with a macro-averaged F1 of 0.40 (loss: 1.2636). The model entirely failed to identify the benign class (precision, recall, and F1 all 0.00; support: 1), while achieving an F1 of 0.80 on the malignant class (precision: 0.67, recall: 1.00; support: 2). This pattern is consistent with a near-degenerate predictor biased toward the majority class.

The texture STL model achieved a test accuracy of 100% and a loss of 0.0018, but a macro-averaged F1 of only 0.33. This apparent contradiction is fully explained by the composition of the test set: all three test samples belonged to the solid class (support: 3/0/0), so a majority-class predictor achieves perfect accuracy while producing zero precision and recall for ground-glass and part-solid. These results are therefore not meaningful and serve only to confirm pipeline correctness.

4.3.1.2 Full Experiment (508 samples)

The full experiment used 508 samples, partitioned into training (306), validation (101), and test (101) sets.

The malignancy STL model achieved a test accuracy of 61.39% and a macro-averaged F1 of 0.56 (loss: 0.6808). Per-class results are reported in Table 4.1. The model demonstrated substantially better recall for the malignant class (0.82) than for the benign class (0.32), consistent with an imbalance-driven bias toward the more frequent positive class.

The texture STL model achieved a test accuracy of 96.04% and a macro-averaged F1 of 0.33 (loss: 0.1671). As in the preliminary run, the macro F1 is severely deflated by the class imbalance in the test set: 97 of 101 samples belong to the solid class, leaving only 2 ground-glass and 2 part-solid samples.

Table 4.1: Per-class metrics for the STL malignancy model, full experiment (held-out test set, $n = 101$).

Class	Precision	Recall	F1	Support
Benign	0.5417	0.3171	0.4000	41
Malignant	0.6364	0.8167	0.7153	60
Macro avg	0.5890	0.5669	0.5577	101

The model correctly classifies all solid nodules (F1: 0.9798) but predicts nothing for the minority classes. Per-class results are reported in Table 4.2.

Table 4.2: Per-class metrics for the STL texture model, full experiment (held-out test set, $n = 101$).

Class	Precision	Recall	F1	Support
Ground-glass	0.0000	0.0000	0.0000	2
Part-solid	0.0000	0.0000	0.0000	2
Solid	0.9604	1.0000	0.9798	97
Macro avg	0.3201	0.3333	0.3266	101

These results establish a task-isolated performance baseline against which the multi-task architectures are evaluated in the following subsection.

4.3.2 Multi-Task Learning (MTL) with Cross-Talk

4.3.2.1 Scope and Limitations

The original design of *MultiTaskCrossTalkNet* targeted three simultaneous tasks: texture classification, malignancy prediction, and longitudinal progression prediction. In practice, the three-task formulation could not be implemented as intended due to a data linkage constraint: the clinical diagnosis file, which contains progression labels, does not share a common patient or nodule identifier with the DICOM XML annotation files from which texture and imaging features are derived. Without a reliable key with which to join these two sources, progression labels could not be unambiguously matched to individual CT crops, making it impossible to include the progression head in any image-driven training pipeline.

Beyond this linkage issue, the progression target is severely class-imbalanced (stable cases substantially outnumbering progressive ones), and predicting longitudinal temporal change from a single static cross-sectional slice presents fundamental modelling challenges. Preliminary tests confirmed that noisy gradients from a progression branch actively degraded the convergence of the other two tasks. The final experiments were therefore limited to the two-task setting: dominant texture category and binary malignancy.

4.3.2.2 Experimental Setup

To prepare inputs for the two-branch networks, a preprocessing pipeline was applied to the raw DICOM files. For each nodule, a representative 2D frame was extracted by selecting the middle slice of the nodule’s spatial sequence using the SOP UID. Pixel arrays were transformed using the DICOM `RescaleSlope` and `RescaleIntercept` attributes, clipped to a Hounsfield Unit (HU) range of $[-1000, 400]$, and normalised to a $[0, 1]$ scale. Images were then cropped around the region-of-interest (ROI) bounding box with a predefined margin and resized to 128×128 pixels.

The dataset was partitioned into training (60%), validation (20%), and test (20%) sets. All MTL variants were trained using the AdamW optimiser to minimise a joint loss: the weighted sum of categorical cross-entropy for the texture head and binary cross-entropy for the malignancy head, with both task weights set to 1.0. The best-performing checkpoint was selected on lowest validation loss and evaluated on the held-out test set.

Two MTL variants were implemented and compared:

- **Cross-stitch:** Each branch exchanges information with the other via a learnable 2×2 mixing matrix inserted after every convolutional block. The matrix linearly combines the full activation tensors of both streams, with weights initialised to favour task-private representations (0.9 on the diagonal, 0.1 off-diagonal) and refined during training.
- **Co-attention:** Each branch derives a channel-wise gate by globally average-pooling the *other* branch’s feature map into a descriptor vector, then passing it through a small squeeze-and-excitation MLP (reduction ratio 8) with a Sigmoid output. The resulting gate is broadcast back to full spatial resolution and used to scale the other branch’s features before adding them as a residual to the current branch. This allows each stream to selectively attend to the channels of the other that are most relevant to its own task, rather than mixing all channels uniformly. As with the cross-stitch unit, co-attention units are inserted after each of the three convolutional blocks, prior to max-pooling.

4.3.2.3 Preliminary Experiment (15 samples)

As with the STL models, a preliminary run on 15 samples was conducted to validate the joint training pipeline. The MTL cross-stitch model achieved a combined test loss of 1.1490. Texture accuracy was 100% (macro F1: 0.33) and malignancy accuracy was 66.67% (macro F1: 0.40). The per-class breakdown mirrors the STL results exactly: the texture head collapses to a solid-only predictor due

to the degenerate test set composition, and the malignancy head fails to identify benign cases. No meaningful comparison between STL and MTL can be drawn at this scale.

4.3.2.4 Full Experiment Results (508 samples)

Results for both MTL variants on the full dataset are reported in Tables 4.3 and 4.4, alongside the STL baseline for direct comparison.

Table 4.3: Texture head performance across all models, full experiment (held-out test set, $n = 101$). STL results repeated from Table 4.2 for comparison.

Model	Class	Precision	Recall	F1	Support
STL	Ground-glass	0.0000	0.0000	0.0000	2
	Part-solid	0.0000	0.0000	0.0000	2
	Solid	0.9604	1.0000	0.9798	97
	<i>Macro avg</i>	0.3201	0.3333	0.3266	101
MTL Cross-stitch	Ground-glass	0.0000	0.0000	0.0000	2
	Part-solid	0.0000	0.0000	0.0000	2
	Solid	0.9604	1.0000	0.9798	97
	<i>Macro avg</i>	0.3201	0.3333	0.3266	101
MTL Co-attention	Ground-glass	0.0000	0.0000	0.0000	2
	Part-solid	0.0000	0.0000	0.0000	2
	Solid	0.9604	1.0000	0.9798	97
	<i>Macro avg</i>	0.3201	0.3333	0.3266	101

Table 4.4: Malignancy head performance across all models, full experiment (held-out test set, $n = 101$). STL results repeated from Table 4.1 for comparison.

Model	Class	Precision	Recall	F1	Support
STL	Benign	0.5417	0.3171	0.4000	41
	Malignant	0.6364	0.8167	0.7153	60
	<i>Macro avg</i>	0.5890	0.5669	0.5577	101
MTL Cross-stitch	Benign	0.6250	0.2439	0.3509	41
	Malignant	0.6353	0.9000	0.7448	60
	<i>Macro avg</i>	0.6301	0.5720	0.5479	101
MTL Co-attention	Benign	0.5000	0.1707	0.2545	41
	Malignant	0.6092	0.8833	0.7211	60
	<i>Macro avg</i>	0.5546	0.5270	0.4878	101

All three models produce identical texture predictions, collapsing entirely to the solid majority class. This outcome reflects the extreme class imbalance in the texture test set (97 solid out of 101 samples) rather than any genuine representational difference between STL and MTL; the cross-talk signal carries no useful information when one task’s supervision is this degenerate.

For malignancy, the picture is more nuanced. The STL model achieves a macro F1 of 0.56, with comparatively balanced benign recall (0.32). The cross-stitch MTL model narrows benign recall further

(0.24) while improving malignant recall (0.90), yielding a slightly lower macro F1 (0.55) but a higher malignant F1 (0.74 vs. 0.72). The co-attention model underperforms both on every malignancy metric, achieving a macro F1 of 0.49. Neither MTL variant offers a clear overall advantage over the STL baseline under these class distribution conditions.

4.3.3 Qualitative Analysis

To complement the aggregate metrics, Figure 4.1 presents four representative nodule crops from the held-out test set, each annotated with its ground-truth label and the prediction produced by the MTL cross-stitch model.

4.4 Discussion

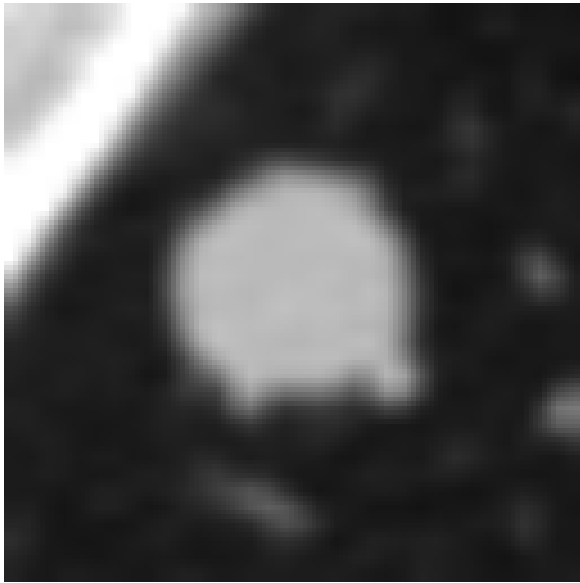
4.4.1 Hypothesis Evaluation

The central hypothesis of this work was that soft parameter sharing via cross-talk between a texture branch and a malignancy branch would improve overall diagnostic performance compared to training each task in isolation. The experimental results do not support this hypothesis under the conditions evaluated.

For texture classification, all three models: STL, MTL cross-stitch, and MTL co-attention produce identical predictions, collapsing entirely to the solid majority class. The macro F1 of 0.33 reported across the board does not reflect a genuine representational equivalence between the models; it reflects the fact that 97 of 101 test samples are solid, making majority-class prediction nearly indistinguishable from high accuracy. The cross-talk mechanism cannot improve texture classification when the supervision signal itself is too imbalanced to provide a learning gradient for minority classes.

For malignancy prediction, the cross-stitch variant achieves a marginally higher malignant-class F1 than STL (0.74 vs. 0.72), but at the cost of substantially lower benign recall (0.24 vs. 0.32), resulting in a slightly lower macro F1 (0.55 vs. 0.56). The co-attention variant performs worse than STL on every metric. The hypothesis that biologically correlated tasks would benefit from shared representations is therefore not confirmed: rather than providing a useful inductive bias, the cross-task signal appears to act as noise that reinforces the model’s existing bias toward the malignant majority class.

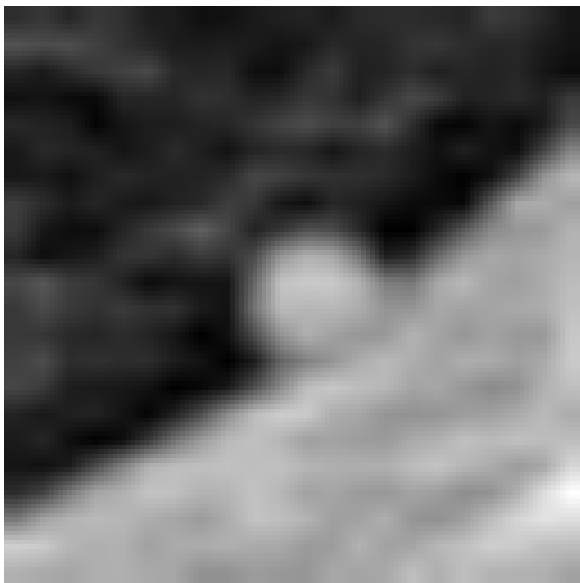
This outcome is consistent with the known risk of negative transfer in MTL, particularly when the auxiliary task (texture) provides a degenerate supervision signal. When the texture branch learns nothing beyond “predict solid always”, the information it passes to the malignancy branch through the cross-talk units carries no useful gradient, and may actively interfere with the more informative



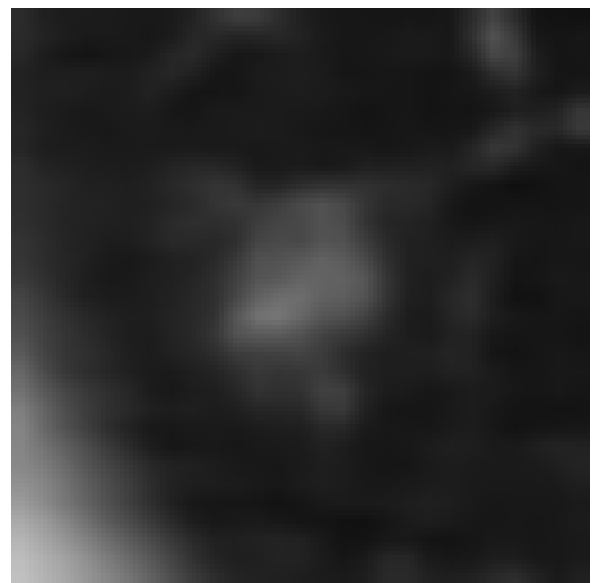
(a) GT: Solid / Benign Pred: Solid / Malignant



(b) GT: Solid / Malignant Pred: Solid / Malignant



(c) GT: Solid / Benign Pred: Solid / Malignant



(d) GT: Ground-Glass / Malignant Pred: Solid / Benign

Figure 4.1: Qualitative results of the proposed model for nodule classification. Each example compares the ground truth (GT) and predicted (Pred) labels for nodule texture and malignancy.

malignancy signal.

4.4.2 Conclusion

This chapter proposed and evaluated *MultiTaskCrossTalkNet*, a multi-task convolutional architecture that jointly predicts nodule texture category and malignancy risk from 2D CT crops via learnable cross-talk between task-specific branches. Two cross-talk mechanisms were compared against a single-task baseline: a cross-stitch unit performing global linear mixing of feature streams, and a co-attention unit performing channel-selective gating based on descriptors from the opposing branch.

Under the experimental conditions evaluated (508 samples drawn from the LIDC-IDRI dataset with severe solid-class dominance) neither MTL variant improved consistently over the STL baseline. The texture task proved unlearnable for minority classes regardless of architecture, and the malignancy task showed no reliable benefit from cross-task information sharing. The primary bottleneck was data quality and class distribution rather than architectural design.

These findings suggest that the value of cross-talk mechanisms in medical imaging MTL is contingent on the quality and balance of supervision available for each task. Future work should prioritize class-balanced sampling strategies, volumetric input representations, and the restoration of the progression task via a corrected data linkage, the condition under which the multi-task hypothesis is most likely to yield measurable gains.

Chapter 5

Conclusion and future work

5.1 Strengths and Limitations

Strengths The experimental design has several genuine strengths. The use of identical branch architectures across STL and MTL ensures that observed differences are attributable to the sharing mechanism rather than to model capacity. The preliminary 15-sample run validated the pipeline before committing to the full dataset, catching degenerate behaviour early. The inclusion of two cross-talk variants: one based on global linear mixing and one on channel-selective gating, provides a meaningful architectural comparison rather than evaluating a single design choice. Finally, the reporting of per-class precision, recall, and support throughout makes the effect of class imbalance transparent and prevents misleading interpretation of accuracy figures.

Limitations The most significant limitation is the extreme class imbalance in the texture task. With 97 of 101 test samples belonging to the solid class, neither the texture classifier nor the multi-task framework can be meaningfully evaluated on ground-glass or part-solid prediction. This imbalance stems from the composition of the LIDC-IDRI dataset itself and from the majority-vote aggregation strategy, which tends to promote the most frequent annotation. Addressing this would require either oversampling of minority-class nodules, class-weighted loss functions, or a substantially larger dataset with better subtype representation.

A second limitation is the reduction from three tasks to two. The exclusion of the progression head due to the missing join key between the clinical CSV and the DICOM XML files removes the task most likely to benefit from shared representations, since longitudinal behaviour is more directly linked to both texture and malignancy than either task is to the other. The architecture was therefore evaluated in a setting weaker than the one it was designed for.

Third, the dataset size of 508 samples is modest for training convolutional networks with multiple independent branches. The co-attention unit introduces additional learnable parameters in the form of MLP gate networks at each depth level, which likely overfit or fail to converge meaningfully at this scale. This may explain why the architecturally richer co-attention variant underperforms the simpler cross-stitch unit despite its greater expressive capacity.

Finally, all models operate on a single 2D slice selected from the middle of the nodule's spatial extent. This representation discards volumetric information that radiologists routinely use to assess nodule morphology. A 3D convolutional or volumetric approach would be more faithful to the clinical diagnostic process, at the cost of substantially higher data and compute requirements.

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